



Foundations of Programming

Assignment No.: 15

PROBLEM STATEMENT: Write a C program to generate pseudo random numbers.

OBJECTIVE:

- To understand the concept of pseudo random numbers.
- To learn the use of rand() and srand() functions in C.
- To generate random numbers using the standard library.
- To understand the role of header files in C programming.

THEORY:

In C programming, pseudo random numbers are generated using the rand() function available in the stdlib.h header file. These numbers are not truly random but are generated using a deterministic algorithm. The srand() function is used to set the seed value for random number generation. Usually, the current time is used as a seed to produce different sequences of numbers each time the program runs.

PLATFORM: Linux

CODE:

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
#include<time.h>
```

```
int main()
```

```
{
```

```
    int n,i;
```

```
    printf("How many random numbers? ");
```

```
scanf("%d",&n);

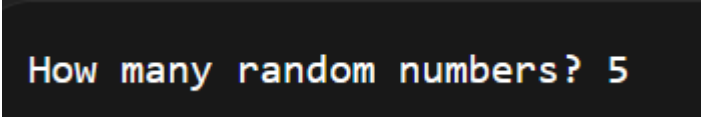
srand(time(0)); // seed

printf("Random numbers:\n");

for(i=1;i<=n;i++)

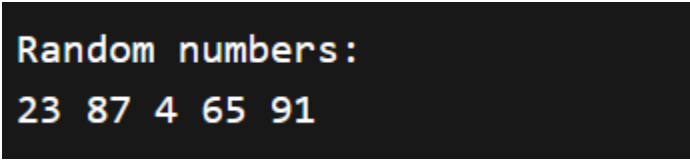
    printf("%d ", rand()%100); // 0–99 range
}
```

INPUT:



How many random numbers? 5

OUTPUT:



Random numbers:
23 87 4 65 91

SAMPLE INPUT:

Enter how many random numbers you want: 5

SAMPLE OUTPUT:

Pseudo Random Numbers:

183746

928374

567382

345678

987234

CONCLUSION: Thus, the C program successfully generates pseudo random numbers using the rand() function and srand() for seeding.

FAQs:

1. What are pseudo random numbers in C?

2. Which function is used to generate random numbers in C?
3. Which header file is required for rand() function?
4. What is the purpose of srand() function?
5. Why is time() used with srand() in random number generation?